



Library Management System

1. Introduction

The Library Management System is a software application developed using Java to simplify and automate the management of library operations. Traditional library systems involve manual record-keeping, which is time-consuming and prone to errors. This project provides a digital solution to manage books efficiently, including adding, viewing, issuing, and returning books. It demonstrates the practical implementation of Object-Oriented Programming (OOP) concepts.

2. Objectives

- To develop a system for managing library records efficiently
- To reduce manual work and human errors
- To implement Object-Oriented Programming concepts in Java
- To provide a simple and user-friendly interface

3. Scope of the Project

This system is designed for small libraries such as schools or colleges. It helps librarians manage book records and track issued books. The system can be further enhanced with database integration and graphical user interface in the future.

4. Features of the System

- Add new books to the system
- View all available books
- Issue books to users
- Return issued books
- Simple menu-driven interface
- Efficient data handling using collections

5. Technologies Used

- Programming Language: Java

- Concept Used: Object-Oriented Programming (OOP)
- Version Control: GitHub

6. OOP Concepts Used

Encapsulation

Data members in classes are kept private and accessed using methods.

Inheritance

Student and Librarian classes inherit properties from the User class.

Abstraction

Abstract class User defines common properties and methods.

Polymorphism

Method overriding is used to define different behaviors.

7. System Architecture

The project follows a layered architecture:

- Model Layer: Book, User classes
- Service Layer: LibraryService class
- Main Layer: Main.java

This structure improves modularity and maintainability.

8. Advantages

- Easy to use
- Reduces manual work
- Efficient data management
- Demonstrates OOP concepts clearly

9. Limitations

- No database integration
- No graphical user interface
- Limited to basic operations

10. Future Enhancements

- Add GUI using Java Swing
- Integrate database (MySQL)
- Add login authentication
- Add search functionality

11. Conclusion

The Library Management System successfully automates basic library operations and demonstrates the use of Object-Oriented Programming concepts in Java. It is a simple, efficient, and scalable solution for managing library activities.